16. Certified energy brackets from reduced-density-matrix relaxations

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Describing every possible configuration of many electrons becomes overwhelmingly expensive as the system grows. Yet the energy of a system with one- and two-particle interactions depends only on certain one-electron and two-electron summaries. The challenge is that not every plausible summary can come from a real many-electron state. We use necessary consistency tests to calculate a guaranteed lower energy limit and an actual trial state to obtain an upper limit.

Key terms

- **Fermion** — A particle, such as an electron, whose quantum states obey antisymmetry under exchange.
- **Reduced density matrix (RDM)** — A summary of selected particles' quantum statistics obtained from a larger system's state.
- **2-RDM** — The two-particle reduced density matrix, sufficient with the appropriate one-particle data to compute the chosen two-body Hamiltonian's energy.
- **N-representability** — The requirement that the reduced data arise from some valid state of N fermions.
- **P, Q and G conditions** — Positivity tests involving particle pairs, hole pairs and particle–hole combinations; they are necessary representability conditions.
- **Hole** — An unoccupied orbital state, treated as an absence of a particle relative to the finite orbital description.
- **Semidefinite program (SDP)** — An optimization problem with linear conditions and positive-semidefinite matrix constraints.
- **Ground-state energy** — The lowest allowed energy of the chosen quantum system.
- **Energy bracket** — A pair of verified lower and upper numbers containing the true ground-state energy.

Why this matters

Reliable energy calculations underpin the theoretical study of bonding and interacting electrons. A bracket tells us how much uncertainty remains, rather than merely reporting a convincing decimal number. The approach may extract useful information without storing a complete many-electron wavefunction, although general representability remains computationally difficult.

What progress would mean

Progress would mean tighter verified brackets at an affordable cost for a specified finite model. That would show how much the extra constraints help; extending the conclusion to an actual molecule would also require controlling orbital-basis and Hamiltonian errors.

Core question

For a specified finite fermionic Hamiltonian, how tight an independently certified ground-state energy interval can P,Q,G and selected higher-order RDM constraints produce?

Scope and assumptions

Fix N≥2 fermions in M spin orbitals with an exact rational/algebraic, number-conserving one- and two-body Hamiltonian. Start with a small Hubbard-type system where exact diagonalization in the same N-particle sector is feasible. Molecular integrals are a later input with an explicit basis and numerical error enclosure.

Use ensemble N-representability. The task is to certify bounds within a finite orbital model, not solve general electronic structure in polynomial time.

Mathematical target

Set $\gamma_{ij} = \langle a_i^\dagger a_j \rangle$ and $\Gamma_{ij,kl} = \langle a_i^\dagger a_j^\dagger a_l a_k \rangle$. Require Hermiticity, fermionic antisymmetry, Tr γ=N, $\sum_{ij} \Gamma_{ij,ij} = N(N-1)$, and $\sum_j \Gamma_{ij,kj} = (N-1)\gamma_{ik}$. Define Q and G using two-hole and particle-hole operator moments and derive their affine forms from anticommutation identities.

Minimizing the linear energy over a necessary-condition outer relaxation gives E_relax≤E_0. A verified dual feasible point gives L≤E_relax; an explicit normalized N-fermion state gives U≥E_0. Report L≤E_0≤U, not an unsupported accuracy estimate from Γ alone.

Required outputs and proof obligations

Export the Hamiltonian and indexing convention, moment constraints, certified dual objective L, state witness for U, and gap U−L. Separate solver error, relaxation error, and finite-basis/model error. Report any property bounds via separately optimized observables; a relaxed Γ need not be physical.

Validation and rejection controls

Compare with exact diagonalization in identical conventions. Check contractions using explicit Slater determinants and correlated states. Adding valid necessary constraints must not lower the exact relaxed minimum. Verify dual PSD matrices and equality residual corrections rigorously, rather than interpreting the raw solver objective as a bound.

Milestones and research extension

Baseline: valid L,U brackets for small fixed models. Strong: quantify improvement from G or T constraints across a specified instance set. Research extension: a new valid constraint with demonstrated tightening, or a theorem on a tractable family; novelty needs comparison with existing hierarchies.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Liu, Christandl and Verstraete, N-representability is QMA-complete — Complexity barrier to general exact representability.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\gamma_{ij} = \langle a_i^\dagger a_j \rangle, \qquad \Gamma_{ij,kl} = \langle a_i^\dagger a_j^\dagger a_l a_k \rangle,$$

$$\operatorname{Tr}\gamma = N, \quad \sum_{ij} \Gamma_{ij,ij} = N(N-1), \quad \sum_{j} \Gamma_{ij,kj} = (N-1)\gamma_{ik}.$$

$$L_{\text{dual}} \le E_{\text{relax}} \le E_0 \le \langle \Psi | H | \Psi \rangle = U, \qquad \langle \Psi | \Psi \rangle = 1.$$

Derive two-particle, two-hole and particle-hole moment matrices from anticommutation in this convention. Necessary PSD constraints give an outer relaxation. Certify dual feasibility for L and a physical N-fermion state for U; a relaxed RDM need not represent any state.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — bound direction (43 versus 106).** Outer RDM relaxations provide lower bounds; physical wavefunctions provide upper bounds. The original states both directions without distinguishing feasible sets.

2. **Blocking — Q definition (26–30, 97–100).** Q is two-hole positivity, not a generic contracted 3-RDM. Its correct affine expression is essential to the SDP.
3. **Major — accuracy certificate (104–109).** A small optimization residual cannot bound distance to the exact ground-state energy without an upper witness or a proved relaxation-error bound.
4. **Major — exponential-wall claim (48–50).** Polynomially many RDM entries do not make representability easy; the general problem is QMA-complete. The revised goal is a rigorous bracket on a declared finite model, with model error separately acknowledged.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a specified finite fermionic Hamiltonian, how tight an independently certified ground-state energy interval can P,Q,G and selected higher-order RDM constraints produce?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: valid L,U brackets for small fixed models. Strong: quantify improvement from G or T constraints across a specified instance set. Research extension: a new valid constraint with demonstrated tightening, or a theorem on a tractable family; novelty needs comparison with existing hierarchies.

Sources supporting the corrections

- Liu, Christandl and Verstraete, N-representability is QMA-complete — Complexity barrier to general exact representability.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.